

Continuous and Discrete Equations for the Current Multi-Ellipsoid Solver

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Abstract

This document records only the equations required for the current production method used in the two-body sweep: an inviscid potential-flow boundary-element solve coupled to a Hamiltonian midpoint step with a coupled endpoint-velocity solve. The method does not use the older pressure-force update, the standalone impulse-difference update, the optional fluid-energy gradient, or the algebraic energy projection. Hydrodynamic energy and impulse are evaluated as state functions of the instantaneous rigid-body state, and each time step solves for endpoint velocities that preserve total linear impulse, total angular impulse, and total kinetic energy.

1 Notation and dimensions

All bold lower- and upper-case symbols denote vectors or matrices. There are N rigid bodies. Body-indexed vectors such as centre $\mathbf{X}_b$, translational velocity $\mathbf{U}_b$, angular velocity $\mathbf{\Omega}_b$, Lamb impulse $\mathbf{L}_b$, angular Lamb impulse $\mathbf{\Lambda}_b$, solid angular momentum $\mathbf{h}_{s,b}$, and unit normal $\mathbf{n}$ all lie in $\mathbb{R}^3$ unless stated otherwise. Unit quaternions $\mathbf{q}_b \in \mathbb{S}^3$ represent orientations.

For the boundary-element discretisation, let M be the total number of surface nodes and let K be the total number of triangular elements over all bodies. The nodal potential vector is $\phi \in \mathbb{R}^M$, the BEM matrix is $\mathbf{A} \in \mathbb{R}^{M \times M}$, and the Neumann right-hand side is $\mathbf{r} \in \mathbb{R}^M$. The stacked endpoint generalized velocity is $\mathbf{z} \in \mathbb{R}^{6N}$; for two bodies, $\mathbf{z} \in \mathbb{R}^{12}$. The conserved impulse vector combines linear and angular parts,

$$\mathbf{C} = \begin{bmatrix} \mathbf{P} \\ \mathbf{H} \end{bmatrix} \in \mathbb{R}^6, \quad (1)$$

where $\mathbf{P} \in \mathbb{R}^3$ is total linear impulse and $\mathbf{H} \in \mathbb{R}^3$ is total angular impulse. The nonlinear endpoint residual is $\mathbf{R} : \mathbb{R}^{6N} \rightarrow \mathbb{R}^7$, so its finite-difference Jacobian is $\mathbf{J} \in \mathbb{R}^{7 \times 6N}$.

2 Rigid-body state

2.1 Continuous variables

Body $b \in \{1, \dots, N\}$ has centre $\mathbf{X}_b(t) \in \mathbb{R}^3$, unit orientation quaternion $\mathbf{q}_b(t) \in \mathbb{S}^3$, semi-axes (a_b, b_b, c_b) , solid density $\rho_{s,b}$, and volume

$$V_b = \frac{4\pi}{3} a_b b_b c_b. \quad (2)$$

The solid mass and body-frame inertia tensor are

$$m_b = \rho_{s,b} V_b, \quad \mathbf{I}_b = \frac{m_b}{5} \begin{bmatrix} b_b^2 + c_b^2 & 0 & 0 \\ 0 & a_b^2 + c_b^2 & 0 \\ 0 & 0 & a_b^2 + b_b^2 \end{bmatrix}. \quad (3)$$

The lab-frame translational and angular velocities are both vectors in $\mathbb{R}^3$:

$$\mathbf{U}_b = \dot{\mathbf{X}}_b, \quad \mathbf{\Omega}_b \in \mathbb{R}^3. \quad (4)$$

The corresponding body-frame angular velocity $\boldsymbol{\omega}_b \in \mathbb{R}^3$ is

$$(0, \boldsymbol{\omega}_b) = \mathbf{q}_b^{-1}(0, \mathbf{\Omega}_b) \mathbf{q}_b. \quad (5)$$

The solid angular momentum $\mathbf{h}_{s,b} \in \mathbb{R}^3$, expressed in the lab frame, is

$$\mathbf{h}_{s,b} = \text{vec} \left[\mathbf{q}_b(0, \mathbf{I}_b \boldsymbol{\omega}_b) \mathbf{q}_b^{-1} \right]. \quad (6)$$

The solid kinetic energy is

$$T_{s,b} = \frac{1}{2} m_b \|\mathbf{U}_b\|^2 + \frac{1}{2} \boldsymbol{\omega}_b^\top \mathbf{I}_b \boldsymbol{\omega}_b. \quad (7)$$

2.2 Discrete state vector

At time t_n the code stores the mixed rigid-body state

$$y_n = \left(\{\mathbf{X}_b^n\}_{b=1}^N, \{\mathbf{U}_b^n\}_{b=1}^N, \{\mathbf{q}_b^n\}_{b=1}^N, \{\mathbf{\Omega}_b^n\}_{b=1}^N \right). \quad (8)$$

The nonlinear solve uses the endpoint generalized velocity vector

$$\mathbf{z} = \begin{bmatrix} \mathbf{U}_1^{n+1} \\ \vdots \\ \mathbf{U}_N^{n+1} \\ \mathbf{\Omega}_1^{n+1} \\ \vdots \\ \mathbf{\Omega}_N^{n+1} \end{bmatrix} \in \mathbb{R}^{6N}. \quad (9)$$

3 Continuous potential-flow problem

3.1 Governing equations

The fluid velocity $\mathbf{u}(\mathbf{x}, t) \in \mathbb{R}^3$ is incompressible, inviscid, and irrotational. Thus

$$\mathbf{u} = \nabla \phi, \quad \nabla^2 \phi = 0 \quad \text{in the exterior fluid domain.} \quad (10)$$

The far-field condition is

$$\nabla \phi(\mathbf{x}) \rightarrow \mathbf{0} \quad \text{as} \quad \|\mathbf{x}\| \rightarrow \infty. \quad (11)$$

On the surface S_b of body b , with outward unit normal $\mathbf{n}(\mathbf{x}) \in \mathbb{R}^3$, the no-through-flow condition gives

$$\frac{\partial \phi}{\partial n}(\mathbf{x}) = \mathbf{n}(\mathbf{x}) \cdot [\mathbf{U}_b + \mathbf{\Omega}_b \times (\mathbf{x} - \mathbf{X}_b)], \quad \mathbf{x} \in S_b. \quad (12)$$

3.2 Boundary-integral equation

The source point is $\mathbf{y} \in S$, and the free-space Green function is

$$G(\mathbf{x}, \mathbf{y}) = \frac{1}{4\pi \|\mathbf{x} - \mathbf{y}\|}. \quad (13)$$

For target point $\mathbf{x}$ on the combined body surface $S = \cup_{b=1}^N S_b$, the direct boundary integral equation is

$$c(\mathbf{x}) \phi(\mathbf{x}) + \int_S \phi(\mathbf{y}) \frac{\partial G(\mathbf{x}, \mathbf{y})}{\partial n_y} dS_y = \int_S G(\mathbf{x}, \mathbf{y}) \frac{\partial \phi}{\partial n}(\mathbf{y}) dS_y. \quad (14)$$

Here $c(\mathbf{x})$ is the boundary solid-angle coefficient. The right-hand side is linear in the rigid-body generalized velocities, but the operator depends on the current body configuration.

4 Continuous energy and impulse state functions

Given the instantaneous solution $\phi : S \rightarrow \mathbb{R}$, the fluid kinetic energy is computed from Green's identity using the exterior-normal convention in the code:

$$T_f = -\frac{\rho_f}{2} \int_S \phi \frac{\partial \phi}{\partial n} dS. \quad (15)$$

The linear Lamb impulse $\mathbf{L}_b \in \mathbb{R}^3$ and angular Lamb impulse $\mathbf{\Lambda}_b \in \mathbb{R}^3$ of body b are

$$\mathbf{L}_b = \rho_f \int_{S_b} \phi \mathbf{n} dS, \quad (16)$$

$$\mathbf{\Lambda}_b = \rho_f \int_{S_b} \phi (\mathbf{x} - \mathbf{X}_b) \times \mathbf{n} dS. \quad (17)$$

These quantities are treated as state functions: no time history of ϕ or finite-difference approximation to $\partial\phi/\partial t$ is used in the current method.

The continuous conserved linear impulse $\mathbf{P} \in \mathbb{R}^3$, angular impulse $\mathbf{H} \in \mathbb{R}^3$, and Hamiltonian $E \in \mathbb{R}$ are

$$\mathbf{P} = \sum_{b=1}^N (m_b \mathbf{U}_b - \mathbf{L}_b), \quad (18)$$

$$\mathbf{H} = \sum_{b=1}^N [\mathbf{X}_b \times (m_b \mathbf{U}_b - \mathbf{L}_b) + \mathbf{h}_{s,b} - \mathbf{\Lambda}_b], \quad (19)$$

$$E = \sum_{b=1}^N T_{s,b} + T_f. \quad (20)$$

5 Discrete boundary-element equations

5.1 Surface discretisation

The combined surface is represented by K six-node quadratic triangular elements. Let ϕ_i be the unknown nodal potential at node $i \in \{1, \dots, M\}$ and let $N_i(\xi, \eta)$ be the quadratic element basis functions. On element e ,

$$\mathbf{x}_e(\xi, \eta) = \sum_{i \in e} N_i(\xi, \eta) \mathbf{x}_i, \quad \phi_e(\xi, \eta) = \sum_{i \in e} N_i(\xi, \eta) \phi_i. \quad (21)$$

The oriented area vector $\mathbf{n} dS \in \mathbb{R}^3$ at a quadrature point is

$$\mathbf{n} dS = \left(\frac{\partial \mathbf{x}_e}{\partial \xi} \times \frac{\partial \mathbf{x}_e}{\partial \eta} \right) \frac{w_q}{2}, \quad (22)$$

where w_q is the triangle quadrature weight.

5.2 Discrete potential solve

The nodal Neumann data $g_i \in \mathbb{R}$ are

$$g_i = \mathbf{n}_i \cdot [\mathbf{U}_{b(i)} + \mathbf{\Omega}_{b(i)} \times (\mathbf{x}_i - \mathbf{X}_{b(i)})], \quad (23)$$

where $b(i)$ is the body owning node i . The BEM discretisation of (14) gives

$$\mathbf{A}(\{\mathbf{X}_b, \mathbf{q}_b\}) \boldsymbol{\phi} = \mathbf{r}(\{\mathbf{X}_b, \mathbf{q}_b\}, \{\mathbf{U}_b, \mathbf{\Omega}_b\}), \quad (24)$$

where $\mathbf{A} \in \mathbb{R}^{M \times M}$, $\boldsymbol{\phi} = (\phi_1, \dots, \phi_M)^\top \in \mathbb{R}^M$, and $\mathbf{r} \in \mathbb{R}^M$. Because the right-hand side is assembled from (23), it is linear in the generalized velocity at fixed configuration.

5.3 Discrete fluid energy and impulses

Let Q_b be the set of element quadrature points owned by body b and $Q = \cup_b Q_b$. At quadrature point q , let $\phi_q \in \mathbb{R}$ be the interpolated potential, $g_q \in \mathbb{R}$ the interpolated normal derivative, $\mathbf{x}_q \in \mathbb{R}^3$ the interpolated position, and

$$\mathbf{a}_q = \left(\frac{\partial \mathbf{x}}{\partial \xi} \times \frac{\partial \mathbf{x}}{\partial \eta} \right)_q \frac{w_q}{2} \quad (25)$$

the oriented area vector. The discrete counterparts of (15)–(17) are

$$T_{f,h} = -\frac{\rho_f}{2} \sum_{q \in Q} \phi_q g_q \|\mathbf{a}_q\|, \quad (26)$$

$$\mathbf{L}_{b,h} = \rho_f \sum_{q \in Q_b} \phi_q \mathbf{a}_q, \quad (27)$$

$$\mathbf{\Lambda}_{b,h} = \rho_f \sum_{q \in Q_b} \phi_q (\mathbf{x}_q - \mathbf{X}_b) \times \mathbf{a}_q. \quad (28)$$

These are the quantities returned by the BEM impulse solve.

6 Discrete conserved quantities

After a BEM solve at a discrete state y , the discrete invariants $\mathbf{P}_h(y) \in \mathbb{R}^3$, $\mathbf{H}_h(y) \in \mathbb{R}^3$, and $E_h(y) \in \mathbb{R}$ are evaluated as

$$\mathbf{P}_h(y) = \sum_{b=1}^N (m_b \mathbf{U}_b - \mathbf{L}_{b,h}), \quad (29)$$

$$\mathbf{H}_h(y) = \sum_{b=1}^N [\mathbf{X}_b \times (m_b \mathbf{U}_b - \mathbf{L}_{b,h}) + \mathbf{h}_{s,b} - \mathbf{\Lambda}_{b,h}], \quad (30)$$

$$E_h(y) = \sum_{b=1}^N T_{s,b} + T_{f,h}. \quad (31)$$

The output columns ‘pcon’ and ‘hcon’ are three-component per-body summands in (29) and (30).

7 Current time step: coupled endpoint-velocity Hamiltonian solve

7.1 Start-of-step targets

At the beginning of substep n , with step size $h > 0$, the solver evaluates the BEM state functions at y_n and defines the target invariant vector

$$\mathbf{C}^* = \begin{bmatrix} \mathbf{P}_h(y_n) \\ \mathbf{H}_h(y_n) \end{bmatrix} \in \mathbb{R}^6. \quad (32)$$

The scalar energy target is the initial total kinetic energy of the run,

$$E^* = E_h(y_0). \quad (33)$$

Using the run initial energy rather than the previous-step energy suppresses long-time accumulated energy drift.

7.2 Endpoint state generated by endpoint velocities

For a candidate endpoint velocity vector $\mathbf{z} \in \mathbb{R}^{6N}$ in (9), the current method constructs the endpoint state by trapezoidal kinematics:

$$\mathbf{X}_b^{n+1}(\mathbf{z}) = \mathbf{X}_b^n + \frac{h}{2} (\mathbf{U}_b^n + \mathbf{U}_b^{n+1}), \quad (34)$$

$$\boldsymbol{\Omega}_{b,m}(\mathbf{z}) = \frac{1}{2} (\boldsymbol{\Omega}_b^n + \boldsymbol{\Omega}_b^{n+1}), \quad (35)$$

$$\mathbf{q}_b^{n+1}(\mathbf{z}) = \exp\left[\frac{h}{2}(0, \boldsymbol{\Omega}_{b,m})\right] \mathbf{q}_b^n. \quad (36)$$

Equation (36) is the quaternion exponential map implemented as

$$\exp\left[\frac{h}{2}(0, \boldsymbol{\Omega})\right] = \left(\cos \frac{\|\boldsymbol{\Omega}\|h}{2}, \frac{\sin(\|\boldsymbol{\Omega}\|h/2)}{\|\boldsymbol{\Omega}\|} \boldsymbol{\Omega}\right). \quad (37)$$

7.3 Endpoint residual

The candidate endpoint state $y_{n+1}(\mathbf{z})$ is pushed into the BEM solver, (24) is solved, and the endpoint invariants (29)–(31) are evaluated. Define

$$\mathbf{C}_h(\mathbf{z}) = \begin{bmatrix} \mathbf{P}_h(y_{n+1}(\mathbf{z})) \\ \mathbf{H}_h(y_{n+1}(\mathbf{z})) \end{bmatrix}. \quad (38)$$

The seven-equation nonlinear residual $\mathbf{R}(\mathbf{z}) \in \mathbb{R}^7$ is

$$\mathbf{R}(\mathbf{z}) = \begin{bmatrix} (\mathbf{P}_h(y_{n+1}(\mathbf{z})) - \mathbf{P}^*)/s_P \\ (\mathbf{H}_h(y_{n+1}(\mathbf{z})) - \mathbf{H}^*)/s_H \\ (E_h(y_{n+1}(\mathbf{z})) - E^*)/s_E \end{bmatrix} \in \mathbb{R}^7, \quad (39)$$

where

$$s_P = \max(\|\mathbf{P}^*\|, 1), \quad s_H = \max(\|\mathbf{H}^*\|, 1), \quad s_E = \max(|E^*|, 1). \quad (40)$$

The accepted endpoint velocity is the candidate returned after iterating toward

$$\mathbf{R}(\mathbf{z}) = \mathbf{0}. \quad (41)$$

For two bodies $\mathbf{z} \in \mathbb{R}^{12}$ and the residual has seven components, so the Newton correction is a minimum-norm correction of an underdetermined system.

8 Nonlinear algebraic solve

8.1 Finite-difference Jacobian

At nonlinear iteration k , the residual Jacobian $\mathbf{J}_k \in \mathbb{R}^{7 \times 6N}$ is approximated by forward differences:

$$\mathbf{J}_{ij}^{(k)} = \frac{R_i(\mathbf{z}_k + \epsilon_j \mathbf{e}_j) - R_i(\mathbf{z}_k)}{\epsilon_j}, \quad \epsilon_j = \epsilon_{\text{fd}} (|z_{k,j}| + 1). \quad (42)$$

Here $\mathbf{e}_j \in \mathbb{R}^{6N}$ is the j th coordinate vector. The current sweep uses $\epsilon_{\text{fd}} = 10^{-3}$ and rebuilds this Jacobian every six nonlinear iterations, with Broyden updates between rebuilds.

8.2 Metric-weighted Newton correction

The residual equations are underdetermined whenever $6N > 7$. The implementation therefore also chooses a correction metric. Let $\mathbf{W}_k \in \mathbb{R}^{6N \times 6N}$ be the symmetric positive correction metric at iteration k . The current default is the block-diagonal solid kinetic metric: for body b , the translational block is $m_b I_3$ and the rotational block is

$$I_{b,\text{lab}}(q_b) = Q_b(q_b) I_{b,\text{body}} Q_b(q_b)^\top, \quad (43)$$

where $m_b \in \mathbb{R}$ is the solid mass, $I_3 \in \mathbb{R}^{3 \times 3}$ is the identity, $I_{b,\text{body}} \in \mathbb{R}^{3 \times 3}$ is the body-frame inertia tensor, and $Q_b(q_b) \in \mathbb{R}^{3 \times 3}$ is the rotation matrix associated with the orientation quaternion q_b . Setting `hamiltonian_coupled_kinetic_metric=0` recovers the old Euclidean choice $\mathbf{W}_k = I_{6N}$.

The code solves the regularised metric-weighted normal equation in residual space:

$$\Delta \mathbf{z}_k = -\mathbf{W}_k^{-1} \mathbf{J}_k^\top \left(\mathbf{J}_k \mathbf{W}_k^{-1} \mathbf{J}_k^\top + \lambda I \right)^{-1} \mathbf{R}(\mathbf{z}_k), \quad \lambda = 10^{-10}. \quad (44)$$

Here $\Delta \mathbf{z}_k \in \mathbb{R}^{6N}$ and $I \in \mathbb{R}^{7 \times 7}$. This is the minimum- $\mathbf{W}_k$ -norm correction that locally reduces the seven invariant residuals:

$$\Delta \mathbf{z}_k = \arg \min_{\Delta \mathbf{z} \in \mathbb{R}^{6N}} \frac{1}{2} \Delta \mathbf{z}^\top \mathbf{W}_k \Delta \mathbf{z} \quad \text{subject to} \quad \mathbf{J}_k \Delta \mathbf{z} = -\mathbf{R}(\mathbf{z}_k), \quad (45)$$

up to the Tikhonov regularisation λ and line search below. The correction is capped as

$$\Delta \mathbf{z}_k \leftarrow \Delta \mathbf{z}_k \min \left(1, \frac{\Delta z_{\max}}{\|\Delta \mathbf{z}_k\|} \right), \quad \Delta z_{\max} = \frac{\delta_{\max}}{h}. \quad (46)$$

The current sweep uses $\delta_{\max} = 0.2$. Backtracking accepts the first scalar $\alpha \in \{1, 1/2, \dots, 1/2^7\}$ satisfying

$$\|\mathbf{R}(\mathbf{z}_k + \alpha \Delta \mathbf{z}_k)\| < \|\mathbf{R}(\mathbf{z}_k)\|. \quad (47)$$

8.3 Broyden update

When enabled, the finite-difference Jacobian $\mathbf{J}_k \in \mathbb{R}^{7 \times 6N}$ is updated after an accepted step by

$$\mathbf{J}_{k+1} = \mathbf{J}_k + \frac{[\mathbf{R}(\mathbf{z}_{k+1}) - \mathbf{R}(\mathbf{z}_k) - \mathbf{J}_k \Delta \mathbf{z}_k] \Delta \mathbf{z}_k^\top}{\Delta \mathbf{z}_k^\top \Delta \mathbf{z}_k}. \quad (48)$$

This is the rank-one secant update used in the current sweep.

9 Adaptive substepping

The nominal step is $h = \Delta t$. If the final coupled residual norm $\|\mathbf{R}\|$ exceeds the scalar tolerance τ

$$\|\mathbf{R}\| > \tau, \quad \tau = \text{hamiltonian_floor_tol}, \quad (49)$$

the whole step is retried with twice as many equal substeps, up to

$$n_{\text{sub}} \leq n_{\text{sub,max}}. \quad (50)$$

In the current sweep, $\tau = 10^{-4}$ and $n_{\text{sub,max}} = 8$. Each substep applies the same endpoint-velocity solve above with $h = \Delta t / n_{\text{sub}}$.

10 Diagnostics

The code reports the physical conservation errors using the unscaled quantities $e_P, e_H, e_E \in \mathbb{R}$:

$$e_P = \|\mathbf{P}_h(y_{n+1}) - \mathbf{P}^\star\|, \quad (51)$$

$$e_H = \|\mathbf{H}_h(y_{n+1}) - \mathbf{H}^\star\|, \quad (52)$$

$$e_E = \frac{|E_h(y_{n+1}) - E^\star|}{\max(|E^\star|, \epsilon_{\text{mach}})}. \quad (53)$$

The CSV output also writes, for each body,

$$\mathbf{p}_{\text{con},b} = m_b \mathbf{U}_b - \mathbf{L}_{b,h}, \quad (54)$$

$$\mathbf{h}_{\text{con},b} = \mathbf{X}_b \times \mathbf{p}_{\text{con},b} + \mathbf{h}_{s,b} - \mathbf{\Lambda}_{b,h}. \quad (55)$$

The sums of these per-body columns are exactly (29) and (30). The study plots use these columns for linear and angular impulse drift, while total kinetic-energy drift is computed from (31).